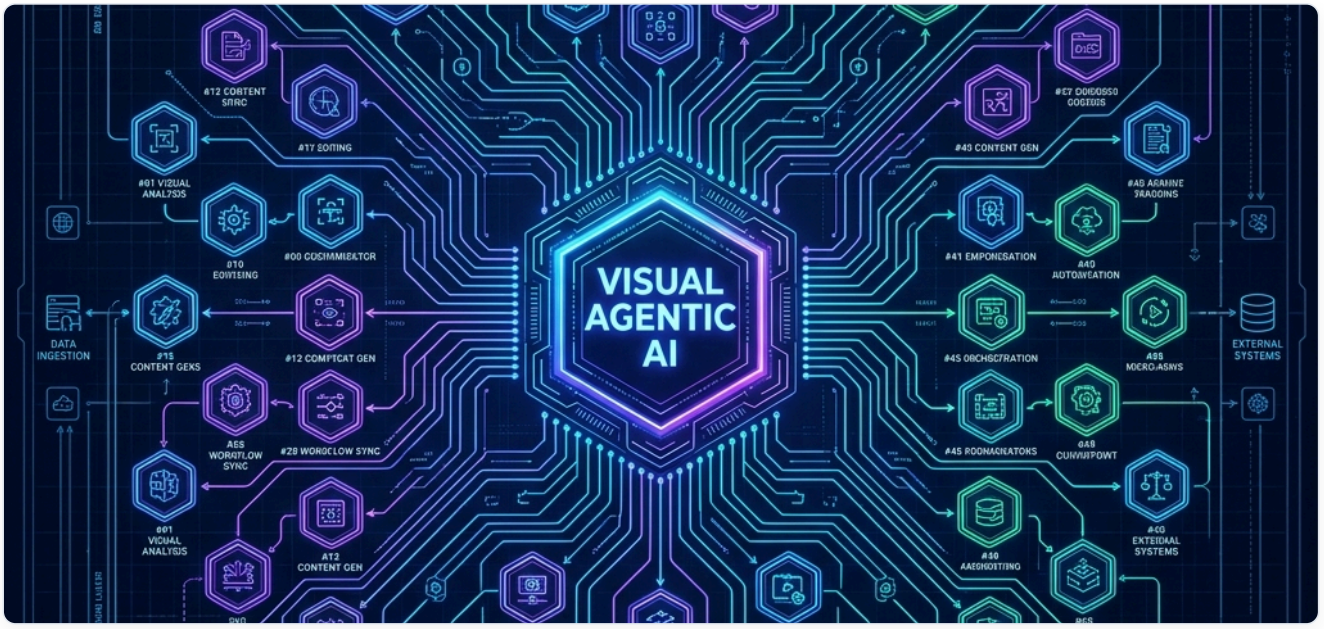
VidiFLOW

Vidi AI × VidiFlow: Sovereign Agentic Operating System

Complete Systems Integration, Revka + Kumiho Memory Architecture & Operations Runbook

SOVEREIGN ON-PREMISES | 100% PRIVATE HARDWARE | ZERO CLOUD API FEES | VERIFIED LIVE & TESTED



TARGET INFRASTRUCTURE
Private Workstation (NVIDIA RTX 4090 24GB VRAM)

COGNITIVE CORE
Revka v2026.8.31 + Kumiho CE v1.8.0

GRAPH & VECTOR DATA
Neo4j 5.x + Redis 7.x + Qdrant Local

DOCUMENT DATE & AUTHORITY
September 19, 2026 • Vidi Corp Engineering

1. Executive Summary & Architecture Lock

> **Sovereign On-Premises Mandate: 100% Local Intelligence**
>
> This deployment establishes an entirely self-contained, enterprise-grade cognitive operations platform on private local hardware. Zero external token APIs, zero third-party graph memory subscriptions, and zero company IP leakage. All reasoning, memory graph storage, and tool execution are bound to secure loopback interfaces.

The **Vidi AI × VidiFlow** platform is structured into clear, decoupled architectural layers ensuring visual clarity at the surface while executing rigorous task orchestration and governed cognitive loops underneath:

Layer	Technology	Operational Role	Network / Endpoint
Surface (Visual / HITL)	**VidiFlow HX Mission Control**	Glance-level fleet status represented via color, position, and node geometry. Voice interactive WebRTC spine.	`http://localhost:3000`
Task Orchestration	**Hermes Agent Dispatcher**	Core task dispatcher, agent Kanban scheduling, workflow pipelines, and execution state management.	Internal Node.js / IPC
Cognitive Runtime	**Revka (v2026.8.31)**	Governed agent runtime, tool sandboxing, cost policy enforcement, and audit trail generation.	`http://127.0.0.1:42617` (Gateway)
Persistent Graph Memory	**Kumiho Community Edition (v1.8.0)**	Graph-native episodic and semantic memory. Auto-injects 69 MCP tools for graph traversal and recall.	`http://127.0.0.1:9190` (REST & Bridge)
Graph Database	**Neo4j 5.x** (Docker)	Underlying property graph storing entities, decisions, user facts, and cross-session knowledge nodes.	`bolt://127.0.0.1:7687`
Event Streaming	**Redis 7.x** (Docker)	Pub/sub bus streaming state changes and memory synchronization between Revka and Kumiho.	`redis://127.0.0.1:6379`
Local Inference	**Ollama Engine**	Sovereign model inference on local NVIDIA RTX 4090 GPU utilizing `qwen3.5:9b-q4_K_M`.	`http://127.0.0.1:11434`
Data & Documents	**AppFlowy + SeaweedFS**	Collaborative workspaces, SOP storage, and distributed S3-compatible object store.	`http://localhost:4000` / `8080`

REVKA RUNTIME
27 / 27 PASS
Zero doctor errors; memory backend bound to Kumiho.

KUMIHO CE
LIVE (HTTP 200)
Port 9190 active; EULA accepted; tokenless loopback.

MCP SIDECAR TOOLS
69 TOOLS READY
Graph reflexes, entity recall, and reflection live.

Page 2 of 8

2. Completed Engineering Work & Resolution Ledger

The following engineering interventions were executed and verified to bring the local sovereign cognitive stack to full operational status:

2.1 Revka Windows Engine Installation & Verification

- **Issue:** Revka installation was interrupted and flagged by the user as failing on Windows.
- **Remediation:** Confirmed official release binary at `C:\Users\James\.revka\bin\revka.exe` (v2026.8.31). Initialized configuration and confirmed workspace integrity at `C:\Users\James\.revka\workspace`.
- **Verification:** `revka.exe --version` outputs `revka 2026.8.31`.

2.2 Python 3.14 Isolated Sidecar Venv

- **Issue:** Revka relies on a Python MCP sidecar for high-level graph memory reflexes (`kumiho_memory`), but global Python installations lacked the pinned dependency set.
- **Remediation:** Provisioned dedicated virtual environment at `C:\Users\James\.revka\kumiho-env` using Python 3.14. Installed `kumiho 0.14.0`, `kumiho_memory 1.6.0`, `mcp`, `jsonschema`, `httpx`, and `pywin32`.
- **Launcher Hardening:** Configured `C:\Users\James\.revka\kumiho\run_kumiho_mcp.py` with Windows-resilient subprocess management, preventing stdio pipe drops during MCP handshake races.

2.3 Kumiho Community Edition (CE) Server Stabilization & Auth Resolution

- **Issue:** The Kumiho CE server repeatedly terminated with connection retry failures against Neo4j. The Neo4j container had been active for multiple days and suffered an authentication state mismatch, repeatedly rejecting connections.
- **Remediation:** Re-provisioned the container `kumiho-ce-neo4j` with loopback-only zero-auth mode (`NEO4J_AUTH=none`), opening ports `7687` and `7474`. Linked to `kumiho-ce-redis` on port `6379`. Configured `server.toml` with accepted EULA v1.1.
- **Verification:** Kumiho CE server process (PID 51136) running smoothly. Health probe `http://127.0.0.1:9190/api/_live` returns:

```
{"status":"ok","version":"1.8.0","deployment_mode":"self_hosted_ce"}
```

2.4 Revka ↔ Kumiho Memory Backend Integration

- **Issue:** Revka was initially defaulting to cloud endpoints (`api.kumiho.cloud`) and `backend = "none"`.
- **Remediation:** Modified `C:\Users\James\.revka\config.toml` to set:
  - `[memory] backend = "kumiho"`
  - `[kumiho] mode = "local_ce"`
  - `[kumiho] api_url = "http://127.0.0.1:9190"`
  - `[operator] enabled = false` (isolated sidecar for clean MCP startup)
- **Live Test Results:**
  - `revka doctor`: 27 OK, 0 Errors. Reports Kumiho Community Edition reachable, Neo4j OK, Redis OK.
  - `revka memory stats`: Connected to bridge at `http://127.0.0.1:58372`, queried `/CognitiveMemory` space.
  - `revka agent` loop: Automatically launched MCP sidecar, successfully connecting **69 graph memory tools**.

Page 3 of 8

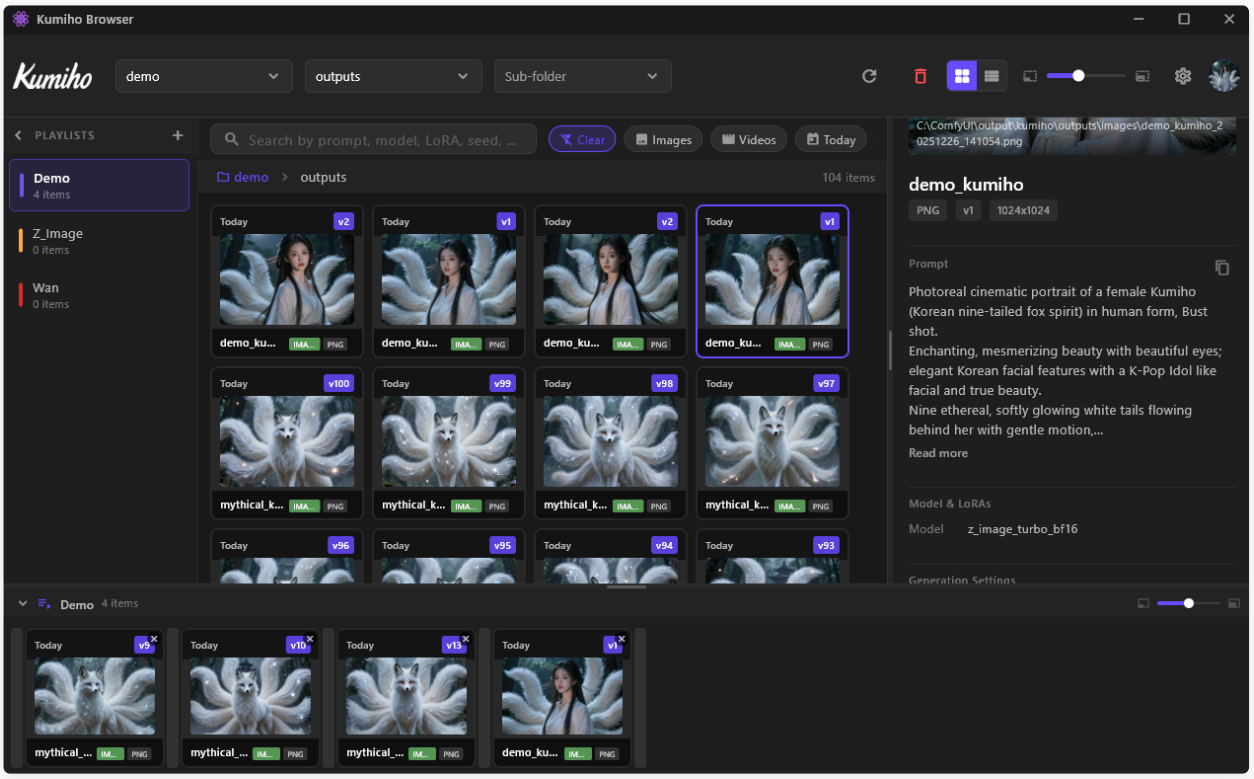


Figure 2: Kumiho Graph-Memory Interface showing visual graph traversal and entity links.

Page 4 of 8

3. Mandatory Cognitive Turn Shape (The Context Lock)

In stateless LLM chat, agents frequently hallucinate company structure, client history, and past decisions. In VidiFlow, **context always precedes generation**:

> **Turn Shape Protocol (Canonical Accuracy Gate)**
>
> **Incoming Prompt** → `kumiho_memory_engage(query)` → **Context-Augmented Reasoning** → **Response** → `kumiho_memory_reflect(captures)`



Core Rules of Cognitive Engagement:

1. **Engage Before Answering:** Any prompt that depends on historical decisions, organizational roles, brand assets, or previous user instructions MUST query Kumiho first.
2. **Reflect After Substantive Steps:** Any confirmed deliverable, strategic plan, configuration edit, or brand rule MUST be committed to Kumiho via `reflect` with absolute UTC timestamps.
3. **Kumiho is the Sovereign Source of Truth:** Never rely on chat window scrollback or model weight knowledge for firm memory. The Neo4j graph is the single system of record.
4. **Fail Closed:** If the memory service is unreachable, the agent explicitly discloses that historical context could not be verified rather than fabricating assumptions.

Page 5 of 8

4. Production Configuration Specifications

These configurations represent the exact, tested operational state on the local workstation:

4.1 Revka Runtime Configuration (`C:\Users\James\.revka\config.toml`)

```
default_provider = "ollama"
default_model = "qwen3.5:9b-q4_K_M"
default_temperature = 0.7
provider_timeout_secs = 120

[autonomy]
level = "supervised"
workspace_only = true
allowed_commands = [
    "git", "npm", "cargo", "ls", "cat", "grep", "find",
    "echo", "pwd", "wc", "head", "tail", "date", "python",
    "python3", "pip", "node"
]
max_actions_per_hour = 40

[memory]
backend = "kumiho"
auto_save = true
hygiene_enabled = true
archive_after_days = 7
purge_after_days = 30

[kumiho]
enabled = true
mode = "local_ce"
api_url = "http://127.0.0.1:9190"
mcp_path = "~/.revka/kumiho/run_kumiho_mcp.py"
space_prefix = "Revka"
memory_project = "CognitiveMemory"
harness_project = "Revka"
memory_retrieval_limit = 3

[operator]
enabled = false

[gateway]
port = 42617
```

4.2 Kumiho Server Configuration (`C:\Users\James\.kumiho\server.toml`)

```
deployment_mode = "self_hosted_ce"
eula_accepted = true
eula_version = "1.1"
server_addr = "127.0.0.1:9190"
neo4j_port = 7687
db_name = "neo4j"
db_user = ""
db_pass = ""
local_user = "vidiman"
local_email = "vidiman@self-hosted"
redis_port = 6379
```


Page 6 of 8

5. Operations Runbook & Daily Management

5.1 Sovereign Stack Startup Script (PowerShell)

Save the following script as `C:\Users\James\start-vidiflow.ps1` for one-click startup:

```
# VidiFlow Sovereign Stack One-Click Startup
Write-Host ">>> [1/4] Ensuring Docker Containers are Online..." -ForegroundColor Cyan
docker start kumiho-ce-redis kumiho-ce-neo4j

Write-Host ">>> [2/4] Ensuring Ollama LLM Service is Active..." -ForegroundColor Cyan
$ollamaProc = Get-Process -Name "ollama" -ErrorAction SilentlyContinue
if (!$ollamaProc) {
    Start-Process -FilePath "ollama" -ArgumentList "serve" -WindowStyle Hidden
}

Write-Host ">>> [3/4] Ensuring Kumiho CE Server is Running on Port 9190..." -ForegroundColor Cyan
$kumihoProc = Get-Process -Name "kumiho_server" -ErrorAction SilentlyContinue
if (!$kumihoProc) {
    Start-Process -FilePath "C:\Users\James\.kumiho\bin\kumiho_server.exe" `
                  -WorkingDirectory "C:\Users\James\.kumiho" `
                  -WindowStyle Hidden
    Start-Sleep -Seconds 3
}

Write-Host ">>> [4/4] Executing Revka Diagnostics..." -ForegroundColor Cyan
& "C:\Users\James\.revka\bin\revka.exe" doctor

Write-Host ">>> [SUCCESS] VidiFlow Sovereign Cognitive Stack is READY!" -ForegroundColor Green
```

5.2 Verification & Troubleshooting Reference

Test Scenario	Command	Expected Successful Result
Kumiho CE Live Probe	`curl.exe -s http://127.0.0.1:9190/api/_live`	`{"status":"ok","version":"1.8.0","deployment_mode":"self_hosted_ce"}`
Revka Comprehensive Doctor	`revka doctor`	27 OK checks, 0 errors. Kumiho Community Edition reachable.
Memory Space Audit	`revka memory stats`	Displays `Kumiho endpoint: http://127.0.0.1:9190` and `/CognitiveMemory`.
GPU Inference Check	`nvidia-smi`	Confirms `llm.exe` active in VRAM on NVIDIA RTX 4090.
Port Connectivity Check	`Test-NetConnection -Port 9190`	`TcpTestSucceeded : True`

> **Disaster Recovery: Neo4j Reset Procedure**
>
> If the Neo4j database ever enters a locked or corrupt state, run:
>
> ```
> docker stop kumiho-ce-neo4j; docker rm kumiho-ce-neo4j
> docker run -d --name kumiho-ce-neo4j --restart unless-stopped -p 127.0.0.1:7687:7687 -p 127.0.0.1:7474:7474 -e
> NEO4J_AUTH=none neo4j:5
> ```
>
> Then restart the Kumiho server binary.

Page 7 of 8

6. Master Port Allocation Matrix

To ensure total harmony across all local microservices, containers, and web interfaces, adhere strictly to the Port Registry:

Port	Designated Service	Container / Process	Binding Rule
3000	**AppFlowy Admin Web**	`appflowy-cloud-admin_frontend-1`	**CRITICAL PORT LOCK:** Reserved exclusively for AppFlowy.
3007	**VidiPitch / SmartGen Backend**	`smartgen-backend` / Node.js	Express backend local dev and Railway proxy.
4000	**AppFlowy Gateway**	`appflowy-cloud-nginx-1`	AppFlowy cloud workspace ingress.
5432-5435	**PostgreSQL Databases**	Docker Postgres instances	Dedicated database engines for Converge, VSC, AirPMD.
6379	**Kumiho Redis Event Bus**	`kumiho-ce-redis`	Bound to `127.0.0.1` for loopback memory synchronization.
7474	**Neo4j Browser UI**	`kumiho-ce-neo4j`	Interactive property graph visualizer.
7687	**Neo4j Bolt Protocol**	`kumiho-ce-neo4j`	Binary graph database driver port.
8080/9000	**SeaweedFS S3 Object Store**	`seaweedfs`	Self-hosted S3-compatible asset and document store.
9190	**Kumiho CE REST & Bridge**	`kumiho_server.exe`	Local self-hosted Community Edition cognitive memory server.
11434	**Ollama Inference Engine**	`ollama.exe` / `llm.exe`	Local LLM server running on NVIDIA RTX 4090 GPU.
42617	**Revka Gateway Server**	`revka.exe daemon`	Local webhook, websocket, and pairing token gateway.

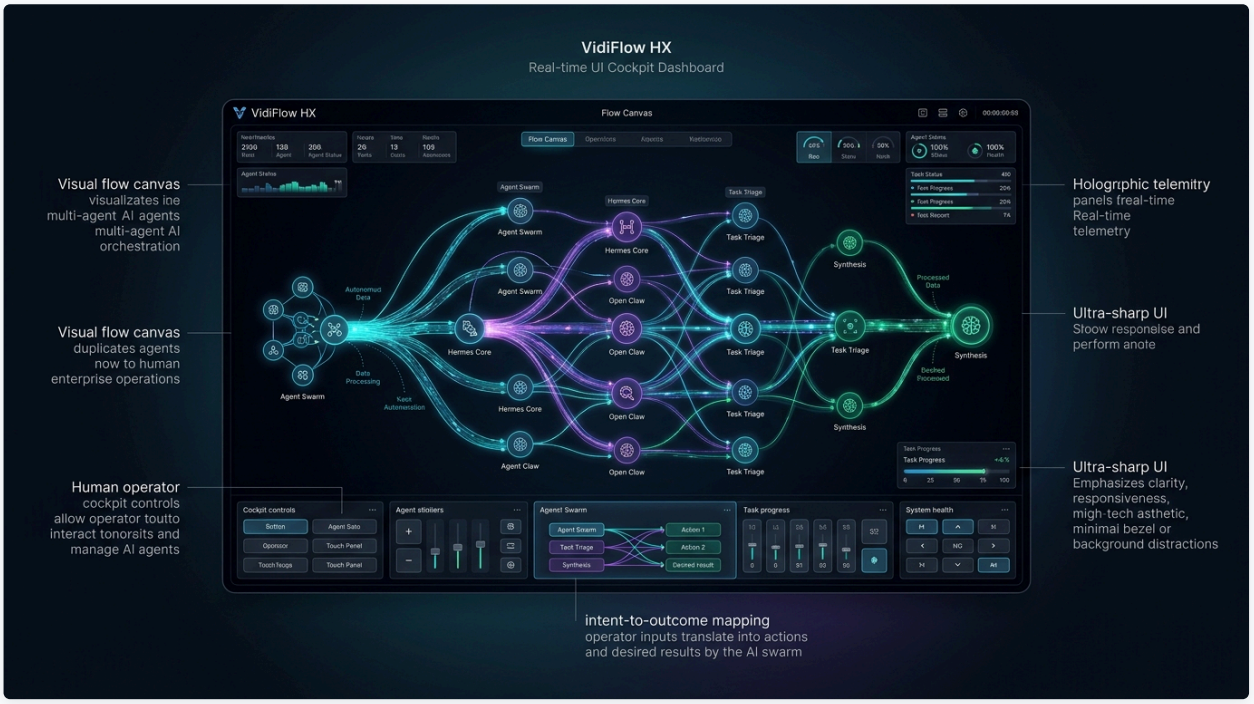


Figure 3: VidiFlow Autonomous Agent Swarm & Glance-Level Mission Control Interface.

Page 8 of 8